Final Project Milestone: Counterfactual Stability Analysis of Classical Lane Perception

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1. Technical Progress

[REPO LINK]

I have implemented an end-to-end classical lane perception pipeline and a counterfactual perturbation framework on KITTI raw sequence `2011_09_26_drive_0002` (77 frames, city driving). The pipeline runs Gaussian blur, Canny edge detection, a tuned trapezoidal ROI mask, probabilistic Hough line detection, slope-based segment grouping into left/right lane hypotheses, and weighted least-squares fitting per group. From the two fitted lines I compute the per-frame vanishing point as the candidate temporal stability signal.

The clean-sequence baseline recovers a vanishing point on **77/77 frames**, with mean frame-to-frame $|\Delta v_x| = 9.55$ px and $\sigma_{v_x} = 24.99$ px. The recovered VP track exhibits a smooth low-frequency drift consistent with ego-motion across the sequence, which establishes that the pipeline produces a coherent temporal signal rather than per-frame noise — a prerequisite for using temporal stability as an uncertainty metric.

I have also implemented one of the three proposed perturbations (synthetic cast shadow) and re-run the full pipeline on the perturbed sequence. The shadow is modeled as a fixed-position dark trapezoid ($\alpha = 0.4$ multiplicative darkening) crossing the mid-distance road surface, simulating a tree shadow. Under perturbation, per-frame VP recovery remains at **77/77**, indicating that any standard frame-level detection-confidence metric would declare the pipeline unaffected. Temporal stability, however, degrades sharply.

2. Visualization & Initial Results

Figure 1: Detected lane boundaries (green) and vanishing point (yellow) on four representative frames of the clean sequence. Ego-motion is visible across the panels.

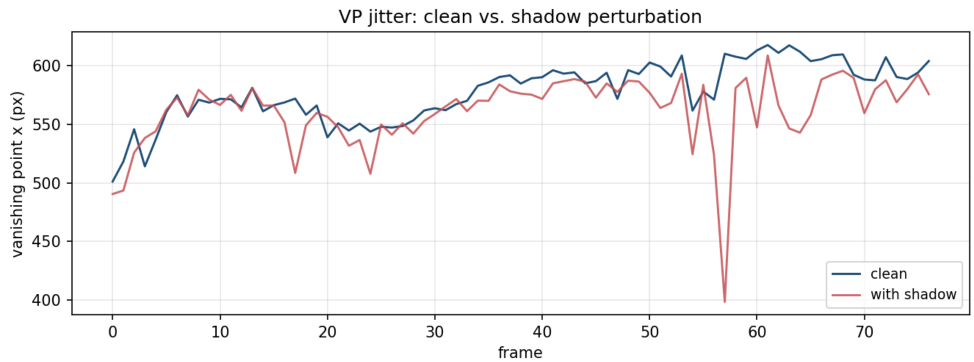

Figure 2: Vanishing point x-coordinate over the full sequence, clean versus shadow-perturbed. Per-frame VP recovery is 100% in both conditions, but the shadow track exhibits a $\sim$200 px excursion at frame 57 and elevated high-frequency variability throughout.

The shadow perturbation increases mean frame-to-frame $|\Delta v_x|$ from 9.55 to 19.30 px (a 2.02$\times$ increase) and σ_{v_x} from 24.99 to 30.03 px. The asymmetric growth (temporal instability scaling more sharply than overall spread) is consistent with the project hypothesis: small semantically plausible perturbations induce temporal incoherence disproportionate to their effect on per-frame outputs. The single-frame excursion at frame 57 is a particularly clear example: a > 200 px VP jump in one frame that would be undetectable by IoU, recovery rate, or any per-frame confidence metric, yet is plainly anomalous when viewed temporally. This supports treating temporal inconsistency as a candidate uncertainty signal.

Two limitations of the current pipeline have been characterized and inform the remaining work. First, the fixed-geometry ROI cannot track ego-vehicle motion within the lane, so some of the clean-baseline drift may reflect ROI mismatch rather than genuine perception jitter. Second, the slope-sign segment grouping detects only the two boundaries of the ego-occupied lane (left dashed centerline and right curb in this sequence) rather than all lane markings, which is appropriate for the trajectory focus but limits multi-lane analysis.

3. Updated Timeline & Plan

The original four-week plan remains feasible with two scope adjustments. Of the three proposed perturbations (lane fading, shadow, motion blur), shadow is complete; I will prioritize motion blur next as it most directly stresses temporal coherence. Lane fading is deprioritized, as is the stretch-goal pretrained-model comparison, in favor of deeper analysis of the classical pipeline's failure modes.

Week 8 (current): Implement motion blur; add temporal stability metrics beyond VP-x (2D VP drift, lane slope oscillation, accumulated heading deviation). Run all metrics across both perturbations and the clean baseline.

Week 9: Implement lane fading if time permits. Analyze which metrics best discriminate perturbed sequences; identify and characterize failure-case frames (e.g., frame 57). Prepare final report, demo, and reproducibility scripts.

Note on AI Assistance:
In preparing this write-up, I used ChatGPT 5.5 as a writing aid and organizational tool. It helped me refine figure captions, adjust section flow, and polish phrasing for clarity, but all substantive content, results, analysis, and reflections are my own. I also used Grammarly to review grammar and tone – suggested edits were selectively incorporated when they aligned with my intended meaning.